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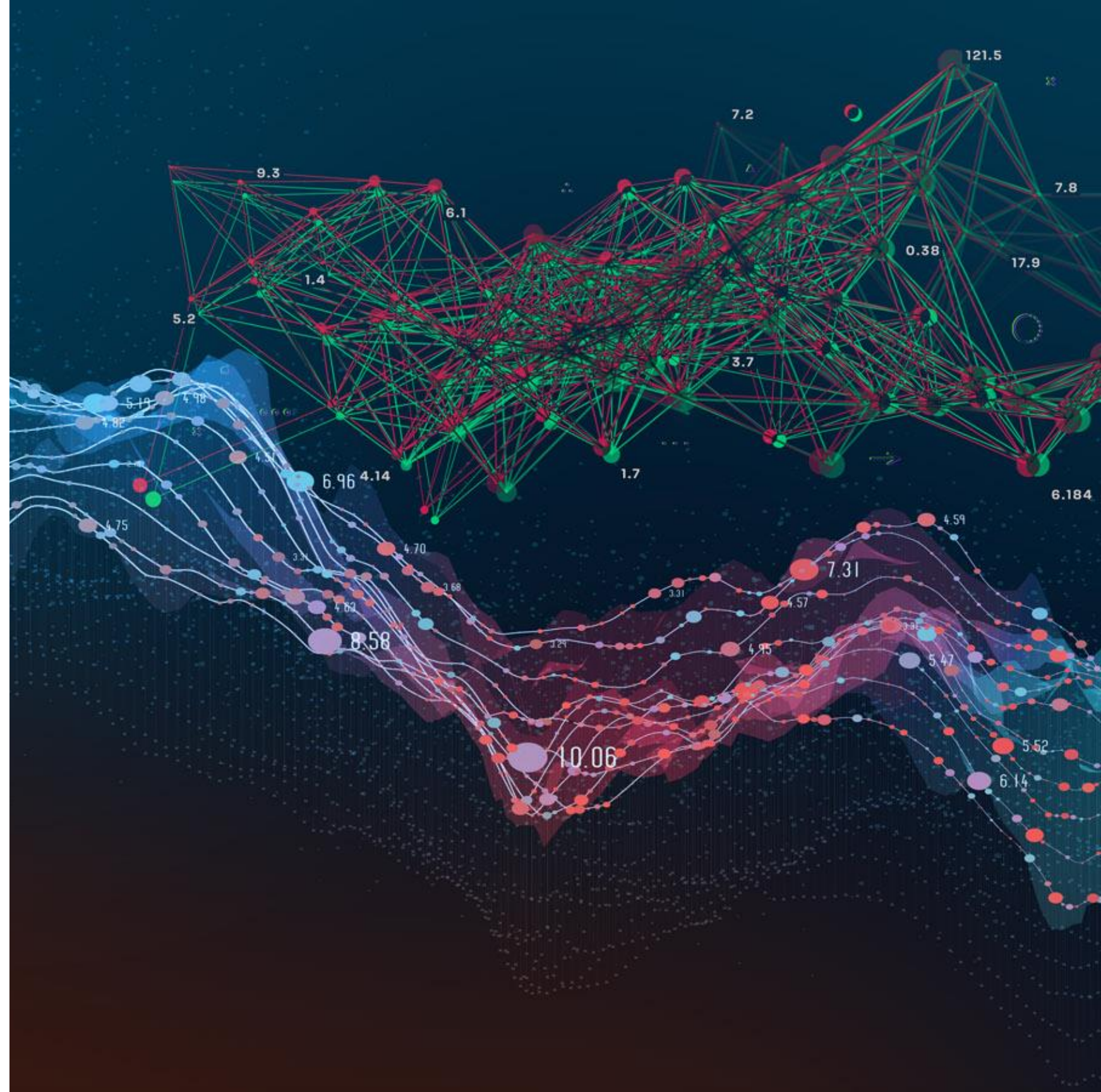


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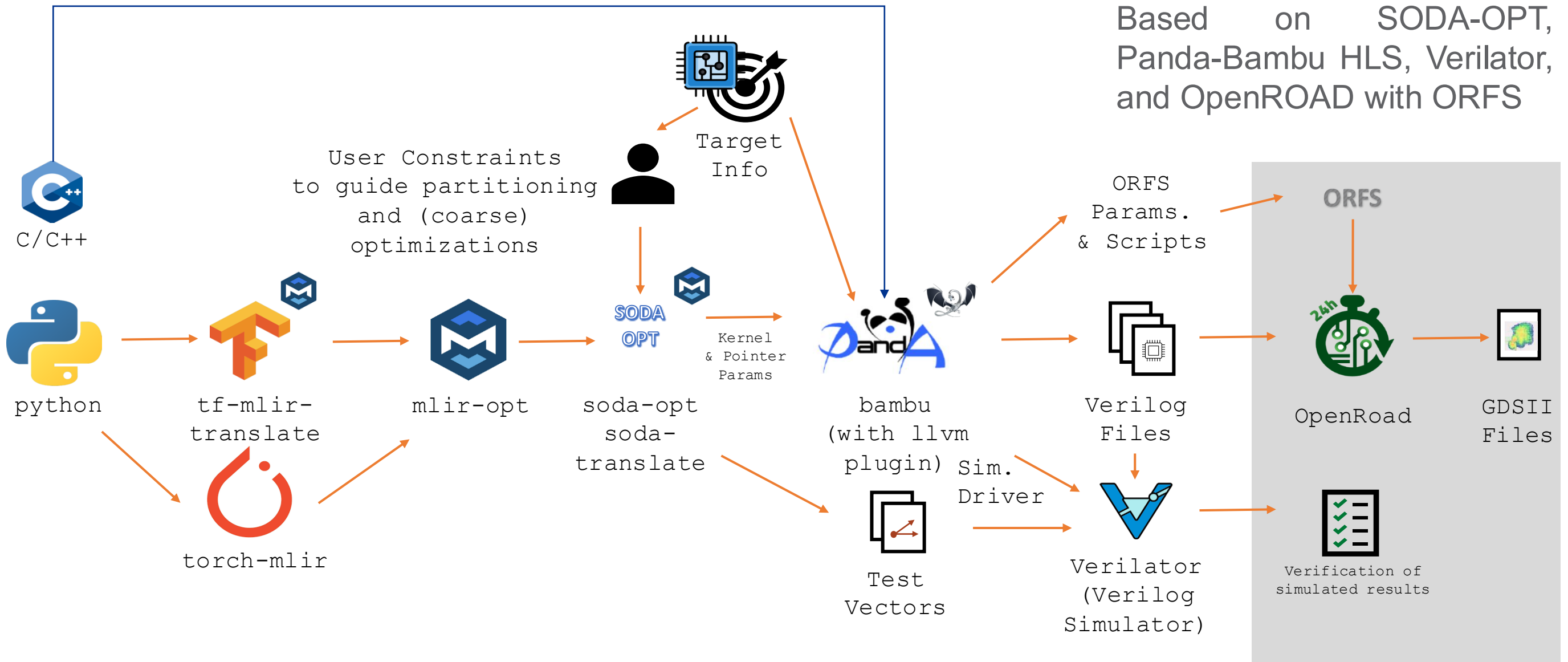


ENCODE (End-to-End Co-design for Performance, Energy Efficiency and Security in AI-enabled Computational Science)

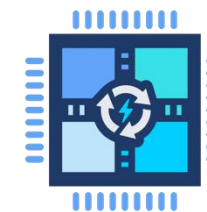


- <https://www.pnnl.gov/projects/encode>
- Develop an integrated end-to-end toolchain of **open-source hardware design tools** for custom **Systems-in-a-Package (SiPs)**
- Develop advanced **hardware synthesis methodologies and tools** to quickly convert high-level specifications and algorithmic formulations into specialized hardware
- Reduce design space exploration and design turnaround time by developing **scalable hardware co-optimization algorithms**
- Establish methodologies and testbeds for **pre- and post-fabrication testing and verification of chiplet-based designs**

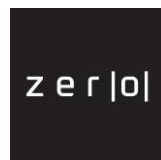
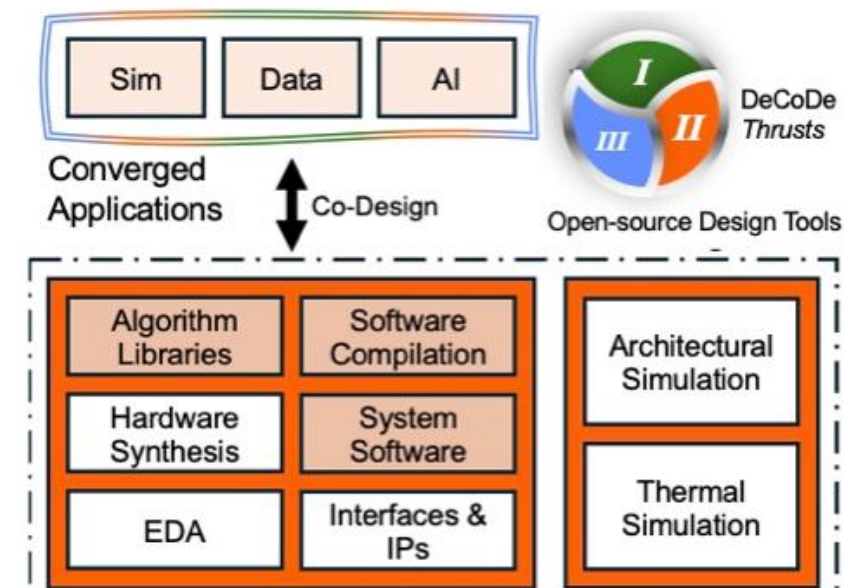
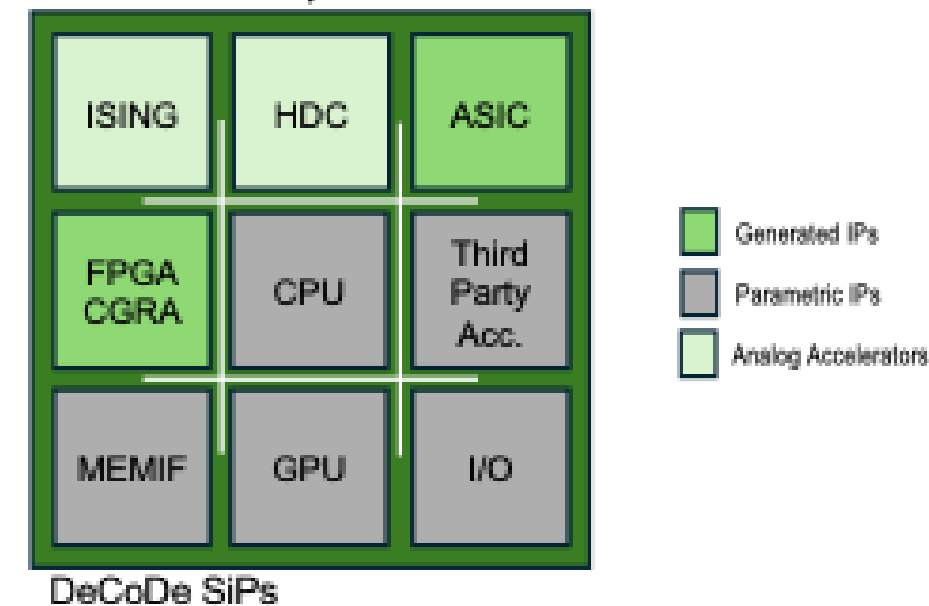
ENCODE First Open-Source End-to-End Pipeline



The Democratization of Co-Design (DeCoDe) for Energy-Efficient Heterogeneous Computing



- <https://www.pnnl.gov/projects/decode>
- Converged workloads drive heterogeneous computing challenges: supercomputers-to-edge with integrated sensors
- Integration of analog accelerators, custom digital accelerators, and commodity CPU/GPUs into SiPs with software stack and algorithms is crucial to reach DOE energy-efficient performance goals.
- Balanced hardware-software co-design methodology, based on open standards, and open-source tools that can lower the cost of hardware Research and Development.
- Provide an open-source agile, low-cost hardware design and generation capability.



An Open Prototyping Capability for Fundamental Research

- DOE-SC requirements for specialized hardware are unique, going from edge devices integrated with sensors, to high-performance computing systems
 - Possible only with new solutions in microelectronics that reimagine co-design, including energy efficient computing and new computing paradigms (see related DOE Reports at: <https://science.osti.gov/ascr/Community-Resources/Program-Documents>)

- We need to complete the open research loop:

Design -> Fabricate -> Evaluate -> Learn -> Share

- This needs:
 - Open, stable, reusable workflows and data sets that lower barriers to entry
 - Access to relevant PDKs, IP, fabrication, packaging, and bring-up
 - Reusable test vehicles, results, and data that magnify community progress
 - Foundation for agentic AI Driven EDA and Design for Prototype
- Call to action:
 - Bring scientific research use cases that cannot cross the prototype barrier
 - Contribute reusable tools, IP blocks, PDK components, benchmarks, or test capability
 - Help define the interfaces that connect the ecosystem

IEEE Computer Magazine Special Issue

- **Democratizing Microelectronics: Open-Source Tools and Chiplet-Enabled Prototyping**
 - Special Issue wants to capture the transformation enabled by Open-Source Hardware Design Tools (from architectural simulators, to generators, to prototyping platforms and electronic Design Automation) and Modular Chiplet-based Design Methodologies
 - Targets prototyping as a way to prove novel hardware designs concepts
- Guest Editors: Antonino Tumeo (PNNL), Andrew Kahng (UCSD), Cristina Silvano (Politecnico di Milano), Jeffrey Voas (NIST)
- Deadline: August 1, 2026
- <https://www.computer.org/digital-library/magazines/co/cfp-democratizing-microelectronics-open-source-chiplet>



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Thank you

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